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Enhancing Well Automation: Pilot Results from Advanced Control Sequences on Several Producer Wells in a Middle East Onshore Field

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Abstract

Middle East onshore upstream facilities typically comprise hundreds of producer and injector wells. Restarting these wells after a full or partial shutdown or simply managing the daily production target under specific process limitations, requires control room operators to adjust control settings on many wells simultaneously. This paper presents the pilot results of an advanced well-automation sequence implemented in the integrated control and safety system (ICSS) of an onshore Middle East field.

The automation sequence includes two control modules. The mono-well module safely starts-up the wells and then regulates them at a specified target, while considering limitations related to equipment, wellbore interface and production policies. The multi-well module monitors liquid, gas and gas-lift process facility status, and triggers corrective actions on each mono-well module when a process limitation is reached. Both modules rely on common sensor signals available in wells and process facilities, while acting on the well chokes and variable speed drives (VSD) when applicable. Remote monitoring of the modules and of the behavior of the wells was done using the available data historian.

The sequence was successfully tested on six gas-lifted wells and three wells equipped with electrical submersible pumps (ESP), located on 2 well pads. After deployment, operators were able to start all wells and bring them to a stabilized state at the defined target, using a limited number of buttons and clicks. Comparative analysis shows that for gas-lifted wells, automated start-up reduced stabilization time by several hours. For ESP wells, it lowered peak discharge pressure and motor current during start-up, helping to minimize equipment stress. Although the observed benefits at field scale remain limited due to the number of wells currently equipped with the solution, the multi-well module was successfully tested, and operators found the solution easier to use compared to manual adjustments.

Integrating such an advanced well control sequence directly into the ICSS offers the advantage of being cyber-proof compared to solutions requiring communication with external controllers and does not incur additional operational costs, such as expensive licenses, once implemented. It also provides greater control over well actions ensuring alignment with best practices and compliance with the recommendations

provided by the Production Optimization team. This approach not only enhances efficiency but also promotes consistency across all wells.

Introduction

Operating artificial lift systems, such as gas-lift or ESP, in upstream facilities that can comprise hundreds of producer wells is a challenging problem that requires the use of advanced control and optimization solutions to make sure that each well consistently operates under optimal control parameter settings. These control parameters typically include the opening of flow control valves or chokes (gas-lift and production lines) and the ESP motor frequency.

The Full Control of Wells (FCW) system is a deterministic control algorithm embedded in the ICSS, first introduced by TotalEnergies in the 90's ([Lemetayer and Miret, 1991](#)), to mitigate production surge during start-up on gas-lifted wells in poorly consolidated west African fields and to minimize the gas-lift used once the well is started. The initial version was implemented on a programmable logic controller (PLC) located on the wellheads platform and would act on each well independently. The benefits of the system were then analyzed in terms of increased well stability and showed to reduce significantly headings phenomena ([Garnaud et al., 1996](#) and [Jansen et al., 1999](#)). It was also improved with the addition of a new module, the 'multi-well', that would be used for sequential restart of a group of wells and for load shedding when some process limitations are reached ([Jansen et al., 1999](#)). The system was later implemented on other brown fields, showing a significant increase in production, up to 10-20% in some cases ([Boisard et al., 2002](#)). Over the years, the FCW logic was refined, deployed across multiple fields and now includes modules for deep offshore wells with riser-based gas-lift, as well as other artificial lift types, including ESP and progressing cavity pump. This paper presents the results of the implementation of the FCW in the ICSS of a Middle East field, for a subset of gas-lifted and ESP wells.

A brief overview of other relevant automation technologies in the case of artificial lift is presented as follows. [Gerrard et al. \(2007\)](#) describes the use of Shell's 'production universe real time optimizer' system, which uses data-driven models to calculate gas-lift flowrate setpoints which are then communicated to the distributed control system (DCS), by the data historian. Operators then validate the new setpoints using a workflow in the DCS. [Campos et al. \(2017a\)](#) and [Campos et al. \(2017b\)](#) present several use-cases – start-up, anti-slugging and gas-lift optimization – using a software called 'module of automated procedure', interfaced with the DCS via an open platform communications (OPC) protocol, where control strategies and heuristics can be implemented. Setpoint selection and activation/deactivation of the controllers are performed by operators using a dedicated human machine interface (HMI) screen in the DCS. More recently, some companies have started to use, on producer wells equipped with artificial lift, control methods typically used in refining and petrochemical plants such as advanced process control (APC). [Patel et al. \(2019\)](#) describes the use of APC for ESP wells while [Hernandez et al. \(2024\)](#) describes the use of model predictive control (MPC), a subset of APC, to regulate and optimize gas-lifted wells at a calculated production flowrate, while considering various limitations such as minimum bottom hole flowing pressure. Finally, the industry is also exploring the use of edge devices, located at the 'edge' (e.g, the wellhead) which are independently analyzing signals, computing optimized control settings and communicating them to the well's actuators ([Liu et al., 2024](#)).

While these new approaches are of great interest, they require the installation of new hardware in the case of edge devices and/or require two-way communication with systems that are external to the ICSS. [Saputelli et al. \(2024\)](#) presents a comprehensive review of digitalization advancements in E&P over the past two decades, emphasizing the growing cybersecurity risks to control systems driven by the increased connectivity. Some of these solutions also require the purchase of commercial licenses which can result in a non-negligible rise in the operating costs of the asset.

The paper is organized as follows. We begin by outlining the design of the FCW solution. Next, we present the main control sequences for both mono-well and multi-well modules and detail a key sub-sequence: the start-up of a gas-lifted well with a closed production choke. Finally, we analyze results from the nine wells tested with the solution, covering system utilization performance indicators from the pilot, comparisons between manual and FCW start-ups for gas-lift and ESP wells, and the application of the multi-well module for production optimization and gas-lift load shedding.

Solution design

The FCW algorithm operates on sequential logic, replicating operator actions through a series of predefined steps triggered by real-time sensor data. It continuously processes real-time data from surface and downhole instrumentation – such as pressure, temperature, flow rate, and ESP motor parameters – through the ICSS. These inputs are used to evaluate the status of the well and initiate transitions between predefined operational steps (e.g., start-up, production, shutdown, load shedding) based on logical conditions. Each FCW sequence is governed by timers, thresholds, and interlocks. For instance, during start-up, the algorithm monitors pressure build-up and flow stabilization. If deviations occur – such as over-pressurization of the production flowline – the FCW automatically initiates corrective actions to prevent safety trips and protect equipment integrity.

For the pilot described, the implementation required the following modifications to the plant hardware and software:

- Valves and VSD ICSS typicals, which refer to control module templates, were updated to accommodate the new control strategies in relation to the FCW logic. Additional HMI screens were developed to provide operators with access to control buttons, parameter inputs and monitoring functionalities.
- Two new controllers were added to avoid overloading the existing ones, each equipped with dual-redundant communication and power supply systems, along with their associated cabling.

Its integration within the ICSS architecture, which also encompasses the emergency shutdown system (ESD), fire & gas (F&G), and process safety system (PSS), is structured to ensure that it does not override or bypass any safety functions, thereby avoiding conflicting actions. This layered approach ensures that the FCW enhances operational efficiency while fully adhering to the established safety philosophy of the site.

The FCW system requires a minimum set of input data from sensors installed at the well. Some sensors are mandatory, while others are ‘recommended’ as they enable additional functionalities when available, like additional production targets or added protections. Mandatory and recommended sensors for gas-lift and ESP wells are given in [Table 1](#). If a mandatory sensor fails, the FCW transitions to a deactivated state without affecting the operating status of the well.

The FCW system acts on a set of remotely operable actuators. These actuators must be functional and responsive, otherwise the system automatically switches to deactivated state to prevent unsafe behavior. The actuators managed by the gas-lift mono-well module are the production choke, the gas-lift choke and the surface safety valves. The actuators managed by the ESP mono-well module are the production choke and the VSD of the ESP. Due to technical constraints, the ESP surface safety valve is excluded from automated control. Control of the actuators is done using proportional integral derivative (PID) controllers, speed ramps, and stepwise adjustments, depending on the variable being regulated. A set of timers is also used for fine-tuning regulation, validating transitions, confirming well states, and minimizing false detections.

Table 1—Mandatory and recommended signals to be used by the FCW for gas-lifted and ESP wells.

	Gas-lifted well	ESP well
Mandatory	Well Head Pressure (WHP) – barg Well Head Temperature (WHT) – °C Casing Head Pressure (CHP) – barg Gas-lift Flowrate (GLFR) – MMscf/d	Well Head Pressure (WHP) – barg Well Head Temperature (WHT) – °C Motor Current (MC) – A Motor Frequency (MF) – Hz
Recommended	Bottom Hole Pressure (BHP) – barg Flowline Pressure (FLP) – barg Production Line Flowrate (PLFR) – stb/d Gas-lift Pressure (GLP) – barg	Pump Intake Pressure (PIP) – barg Flowline Pressure (FLP) – barg Production Line Flowrate (PLFR) – stb/d Current Leakage – mA Cable Fault Resistance – kΩ Pump Backspin Relay – Boolean Pump Discharge Pressure (PDP) – barg Pump Discharge Temperature (PDT) – °C Motor Winding Temperature – °C Motor Vibration – G

To enable operators to monitor FCW actions on the wells, dedicated ICSS HMI screens were configured. For remote, office-based monitoring, all FCW tags (signals) were also created and integrated with a commercial data historian, with dedicated dashboards.

Control sequence description

The FCW system integrates two distinct but complementary control modules: the mono-well and the multi-well. These modules operate within a unified supervisory framework, enabling seamless transitions and coordinated control across multiple wells.

Mono-well

The mono-well module governs the autonomous control of a single well. Input parameters such as targets and limitations are configured through the HMI. It can operate independently or together with the multi-well module.

Main sequence. The main sequence governing the mono-well module is detailed in [Figure 1](#). The FCW system operates through a structured sequence of control states. These states – deactivated, progressive stop, ready to start, start-up, production, and temporary stop – are interconnected through defined transitions, ensuring safe and efficient well management under both normal and degraded conditions.

The logic begins in the deactivated state, where the FCW is idle and all control logic is suspended. This state serves as the baseline for safety and maintenance operations. Upon activation, all permissive conditions are verified, including actuator readiness and sensor validity, before proceeding. From ‘deactivated’, the system can transition to the ‘ready to start’, ‘progressive stop’ or ‘production’ steps triggered by an operator clicking on the corresponding button.

Once the system reaches the ‘ready to start’ step, it transitions into the ‘start-up’ state when it receives a start-up command. In this phase, the FCW gradually ramps up the well’s production by adjusting the actuator positions and the motor frequency (for ESP wells) to stabilize flow conditions and avoid abrupt changes in operating conditions.

Following successful stabilization, the system transitions into the ‘production’ state. This is the core operational state where the FCW continuously regulates production parameters to meet defined operating targets. It dynamically adjusts control valves and motor speeds based on real-time sensor feedback and applies protective logic to mitigate risks such as ESP gas lock, reservoir management, well and equipment protection. The system remains in this state until a stop command or fault condition is detected, or an FCW OFF command is given.

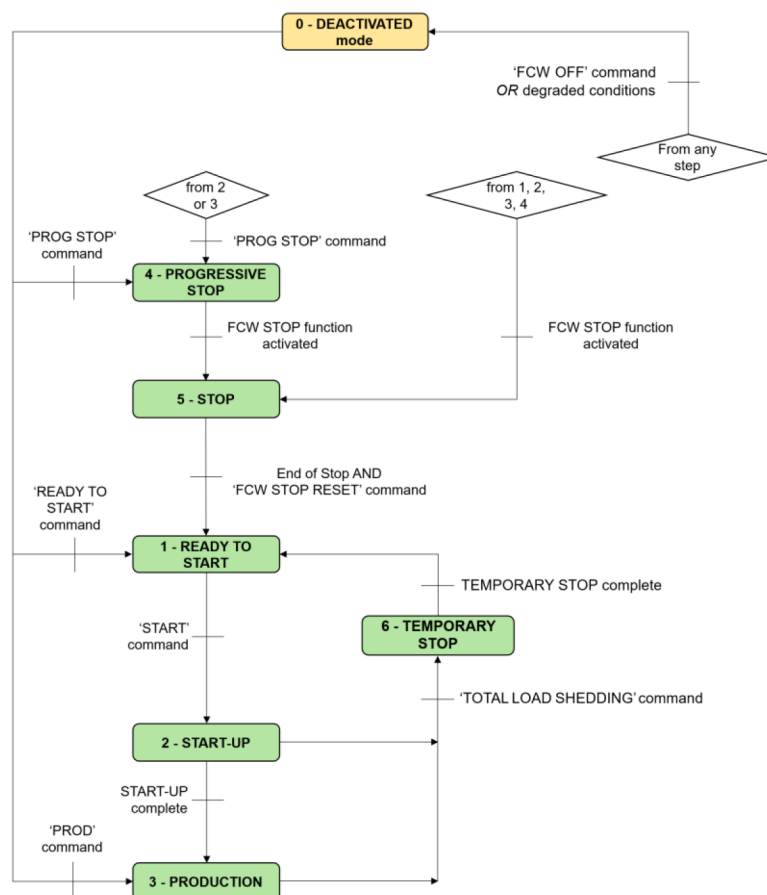


Figure 1—Mono-well sequence chart. Blocks indicate system states and arrows show transitions triggered by specific conditions. Commands correspond to operator buttons on the HMI.

When a controlled reduction in production is required, a well can be stopped using the ‘progressive stop’ sequence. In this phase, the FCW gradually reduces production by closing the chokes and decreasing the motor frequency (for ESP wells) until the well is completely stopped. This smooth and controlled ramp-down prevents pressure surges in the well borehole area and allows a progressive reduction of the operating regime of the ESP to an idle mode prior shutdown.

Finally, if a temporary production interruption is triggered by the multi-well module, due to process limitations or manual activation of a load shedding scenario, the system enters the ‘temporary stop’ state. In this step, the FCW executes a controlled ramp-down of production and remains on hold until the process limitations are cleared to resume well production.

As described, the FCW logic manages well operations for start-up, shutdown, and regulation at the specified target. It also contributes to the protection of the well through some specific functions. Both gas-lift and ESP mono-well modules include the following functions:

- Backflow protection function: the delta pressure (dP) across the production choke valve is monitored. When the dP is lower than a defined threshold, the production choke is gradually closed to restore a minimal dP and thus prevent backflow from occurring on a producer well.
- Reservoir wellbore protection function: the absolute pressure (BHP / PIP / WHP), temperature (WHT) and the pressure variations are continuously monitored and compared to predefined thresholds. Exceeding these thresholds triggers a ramp-down in response to high absolute pressure or temperature, and triggers a pause action in the case of abrupt pressure variation, thereby protecting the wellbore from damage due to excessive drawdown variations.

Gas-lift sequence specificities. The gas-lift mono-well module includes a flowline pressure protection function. If the pressure downstream the production choke exceeds a defined limit, the well opening is paused until the production kick fades away. Once the pressure normalizes, the FCW will resume the ramp-up of the well. By setting the limit below the safety threshold, a safety trip of the well can be avoided and consequently, associated production shortfalls are reduced.

The gas-lift module includes several types of possible start-up for a well. Figure 2 shows the sequence for starting a gas-lifted well with a closed tubing. Similar logic charts are available for other types of gas-lift start-up, for ESP start-up and for the normal production state; however, these are beyond the scope of this paper and are not described herein. The closed tubing start-up steps are described as follows:

- Step 1: the logic initiates the ‘closed’ start-up sequence.
- Step 2: the production choke and gas-lift control valve are closed. These settings prepare the system for the opening of the isolation valves.
- Step 3: the logic then opens the safety valves on both the production and gas-lift lines. Once the system receives confirmation that the valves are open, it proceeds to the next step.
- Step 4: since the manual wing valves are not equipped with position feedback, their opening must be confirmed by the operator through an ‘acknowledge’ button.
- Step 5: the system waits for the gas-lift network pressure to reach a minimum threshold. This step typically relies on a sensor that, while not essential to the operation of the tool, adds value in terms of operational monitoring. If the sensor is unavailable, the logic can still proceed based on operator acknowledgement.
- Step 6: the casing is pressurized to an intermediate pressure, following a maximum ramp rate to preserve the integrity of the gas-lift valves when the casing is filled with liquid.
- Step 7: the casing is further pressurized to its kick-off pressure using a specific ramp rate. Once this target is reached, the logic transitions to the next phase.
- Steps 8 and 9 (in parallel): on the gas-lift side, the regulation mode switches from pressure to flow control to achieve the desired kick-off rate. Simultaneously, the production choke begins to open gradually until stable operating conditions are reached. Once both objectives are met, the logic proceeds to the final step.
- Step 10: the gas-lift valve is regulated to maintain the kick-off gas-lift flow rate, while the production choke is also controlled to ensure stable operating conditions. This state is maintained for a predefined duration to confirm system stability. After the timer elapses, the logic transitions to the ‘production’ state.

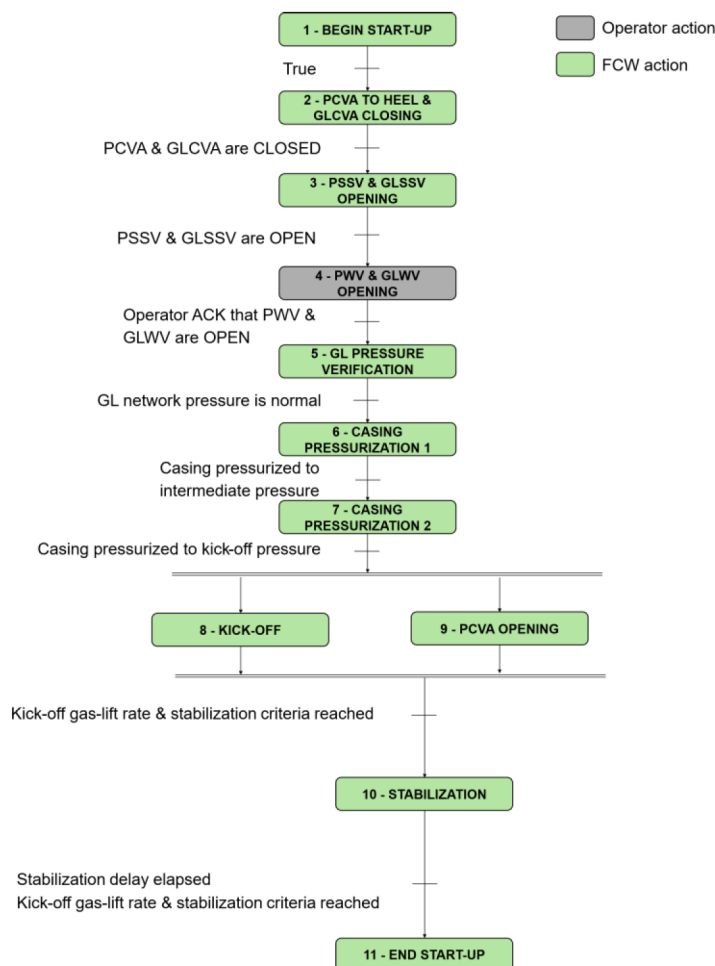


Figure 2—Mono-well ‘closed’ start-up sequence chart of a gas-lifted well. Blocks indicate system states and arrows show transitions triggered by specific conditions.

ESP sequence specificities. The ESP mono-well module includes specific functions to protect the pump:

- **ESP fault function:** it monitors the ESP signals and compares them to defined thresholds. The logic aims to prevent safety trip of the well through early detection of deviations compared to the ESP normal operating conditions. Depending on the severity of the deviation, the logic provides a gradual response. As a first step, the ESP pump frequency is reduced to a fallback position to restore normal operating conditions and provide sufficient time for the operations team to investigate the root cause without immediately stopping production. If the severity of the deviation increases and one of the protection variables is triggered, the logic initiates ESP stop, protecting the pump. Once the issue is acknowledged by the operations team, the well can be restarted remotely using the FCW without external intervention.
- **Gas-lock function:** it monitors any gas lock phenomena that can severely damage the pump. The motor current is monitored and when a sudden drop or an instability is observed, the logic takes action to reduce the quantity of fluid produced and pressurize the tubing from the upstream production choke down to the reservoir in order to reduce the gas-liquid ratio going through the pump, thus restoring normal operating conditions.

Multi-well

The multi-well module orchestrates the coordinated control of several ‘mono-wells’ connected to a shared production system. It essentially performs two actions: it automates the initiation of the mono-well start-up

sequences, and it triggers automatically load-shedding strategies based on process limitations. Commands are relayed to individual wells through their respective mono-well modules. Load-shedding scenarios are designed to respond to process constraints, such as separator capacity, compressor availability or flaring constraints. The multi-well module executes well operations in a sequential manner using a start-up and load-shedding table that defines the order. It supervises global parameters and only controls wells operating in multi-well mode. Figure 3 shows the sequence governing the multi-well module.

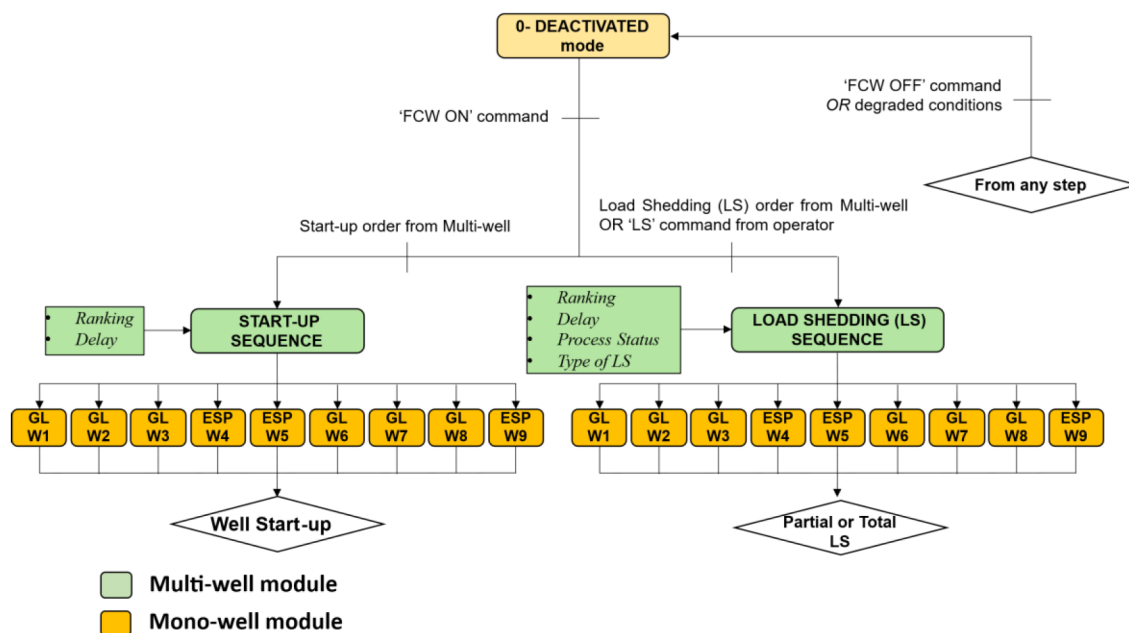


Figure 3—Multi-well sequence chart. Rounded blocks indicate system states, sharp-edged blocks denote input parameters, and arrows indicate transitions triggered by specific conditions.

The multi-well module manages the start-up of multiple wells based on a configurable ranking table. Each well is assigned a rank (1, 2, 3, etc.), determining its start-up order. A rank of 0 disables the automatic start-up. Delays can be added between well startups to allow pressure and flow stabilization on the process side. The system also verifies that production line pressure is below a threshold before initiating each startup.

The load shedding functionality involves continuous monitoring of process constraints, with automatic activation of load shedding when thresholds are exceeded. The systems monitored by the multi-well are the first stage production separator levels, the produced water tank separator levels, the flared flowrates and the gas-lift network pressure (only applicable for gas-lift wells). Each system has its own ranking table. These tables define the order of well curtailment, the expected actions, and any delays between successive load-shedding events.

A load shedding scenario is either triggered automatically (result of process constraints reached) or manually by an operator. Load shedding can result in three possible outcomes for each well:

- No impact (well not in multi-well mode or excluded by rank);
- Partial reduction (production curtailment);
- Full shutdown (smooth ramp-down to closure).

The automatic triggering conditions are defined as follows. Each system is categorized as ‘normal’ when no threshold is exceeded, ‘high/low’ when an intermediate threshold is exceeded, and ‘very high/very low’ when an upper threshold is surpassed. These dual threshold levels allow for the design of customized scenarios and proportionate actions on wells. For example, a well may initially undergo partial

load shedding, and if conditions escalate, it can then be fully load-shed. As for the start-up ranking table, a delay between the load shedding of two wells can be set to help stabilize the process and to ensure smooth flow conditions while the logic performs the necessary adjustments. Once process conditions are normalized, or following the deactivation of the load shedding command, manually by the operator, the wells either ramp up back to their initial production targets, if they were partially load-shed, or restart automatically if they were fully load-shed. This entire sequence is autonomously managed by the control logic.

Pilot implementation

Operating philosophy prior to FCW implementation

The infrastructure of the onshore production site comprises numerous well pads and off-pad wells, interconnected via flowlines to intermediate gathering stations, and subsequently to a central degassing station (CDS). The CDS also receives the production from an adjacent unmanned satellite field, which is operated remotely from the same control room. Combined, the two fields comprise over 200 flowing producer, water and gas injector wells, demanding significant effort to ensure safe and efficient management.

Before FCW implementation, gas-lift activation for the selected wells was controlled remotely but remained a manual process. No feedback loop was implemented in the DCS to regulate the gas-lift choke, meaning valve positions were adjusted based on support team instructions. During well kick-off phases, casing pressurization thresholds and expected gas-lift flow rates were largely determined by the experience of panel operators. As the valves were manually controlled, pressure variations in the gas-lift network could not be dynamically accounted for. Although currently marginal, unstable operating conditions could intensify as the water-cut increases. This operational instability leads to suboptimal utilization of lift gas, reduced production efficiency, and potential disturbances in surface processing facilities further complicating production operation.

Similarly, for wells equipped with ESPs, the production choke valve was manually adjusted from the control room to maintain a target well head pressure. This static approach limits the system's ability to adapt in real time to fluctuations in flowline pressure and, without continuous operator oversight, could result in unstable flow rates. These instabilities not only impact neighboring wells within the same manifold but also increase the likelihood of operating ESPs outside their optimal performance range and safe operating envelope. The methodology for calculating the ESP operating envelope and strategies to ensure operation within it are detailed in [Al Mawali et al. \(2024\)](#).

Well architecture and simplified process used

[Figure 4](#) depicts simplified schematics of the wells selected for the pilot, covering both gas-lifted and ESP-equipped, along with the available signals, actuators and controllable operating targets available through the FCW system.

For gas-lifted wells, the production choke can be operated under three control strategies: fixed opening (PCVA), constant differential pressure across the choke (dP PCVA) or constant well head temperature (WHT). The gas-lift choke is regulated to maintain a constant gas-lift flow rate (GLFR). For ESP wells, the motor frequency can be controlled to achieve a specific well head temperature (WHT), pump intake pressure (PIP) or a fixed value (MF), while the production choke is controlled to maintain a constant well head pressure (WHP). During the pilot, FCW input parameters were selected by running optimization workflows using well-test calibrated steady-state well models, complemented by empirical tuning through trial and error on individual wells.

A simplified process layout used to configure the multi-well module during the pilot is shown in [Figure 5](#). The six gas-lifted wells receive injection from a shared gas-lift manifold while their production is routed with ESP wells to a common first stage separator. After separation, water is routed to storage tanks, while

Results and performance analysis

This section presents selected results from the application of FCW over several months of operation. A review of system utilization is first presented. We then illustrate the use of the system by comparing gas-lift and ESP well stop and restart, with and without FCW. Finally, the functionality of the multi-well module is demonstrated through the activation of two partial load-shedding scenarios – one for production and one for gas-lift – and their respective impact on the wells' behavior.

System utilization

Table 2 presents the key performance indicators (KPI) calculated for the pilot to assess operator utilization of the FCW system following its commissioning. These KPIs include: the percentage of time FCW was 'on', the percentage of time it remained 'on' while the well was flowing, and the proportion of start-up and shut-in executed using FCW. The analysis was limited to periods when the historian system was operational, as server connectivity issues occasionally interrupted data logging. This limitation is reflected in the corresponding row of the table.

Table 2—Key performance indicators on system utilization

KPI	GL Well 1	GL Well 2	GL Well 3	ESP Well 4	ESP Well 5	GL Well 6	GL Well 7	GL Well 8	ESP Well 9	Total
% of time FCW is on	74%	83%	90%	99%	78%	72%	66%	78%	49%	77%
% of time FCW is on & well flowing	75%	84%	91%	99%	73%	74%	70%	78%	88%	82%
% of start-up using FCW	58%	67%	45%	100%	57%	60%	37%	60%	67%	56%
% of stop using FCW	33%	50%	27%	100%	85%	60%	22%	60%	66%	47%
Nbr of stops & start-up	12	6	11	2	7	5	8	5	3	59
% of time PI historian is on	97%	95%	96%	96%	97%	96%	83%	93%	79%	93%

During the monitored timeframe, FCW was active for more than 75% of the total time, and more than 80% of the time when the well was flowing. Periods of inactivity were attributed to several factors. Technically, failures such as hydraulic pump malfunction – responsible for maintaining control line pressure and keeping the surface safety valve (SSSV) open – led to automatic FCW deactivation. Operationally, FCW was sometimes deliberately turned off for ESP wells with high water-cut while closing them to support water management strategies. Additionally, change management considerations played a role, as operators required time to build confidence in the system and fully integrate it into their workflows.

Start-up operations showed stronger adoption of the tool, with 56% executed via FCW. In contrast, only 47% of shut-ins were performed using the system, again largely due to change management dynamics and operator familiarity. It is worth noting that some wells experienced a high number of stops and restarts, exceeding 10 occurrences, primarily due to pigging operations that require temporary reductions in production.

Stop and start-up of a gas-lifted well

An example of a typical progressive stop followed by a ‘closed’ start-up (sequence detailed in Figure 2) of a gas-lifted well, using FCW, is shown in Figure 6. For this graph and all subsequent graphs, well signals were sampled at a rate of one point per minute.

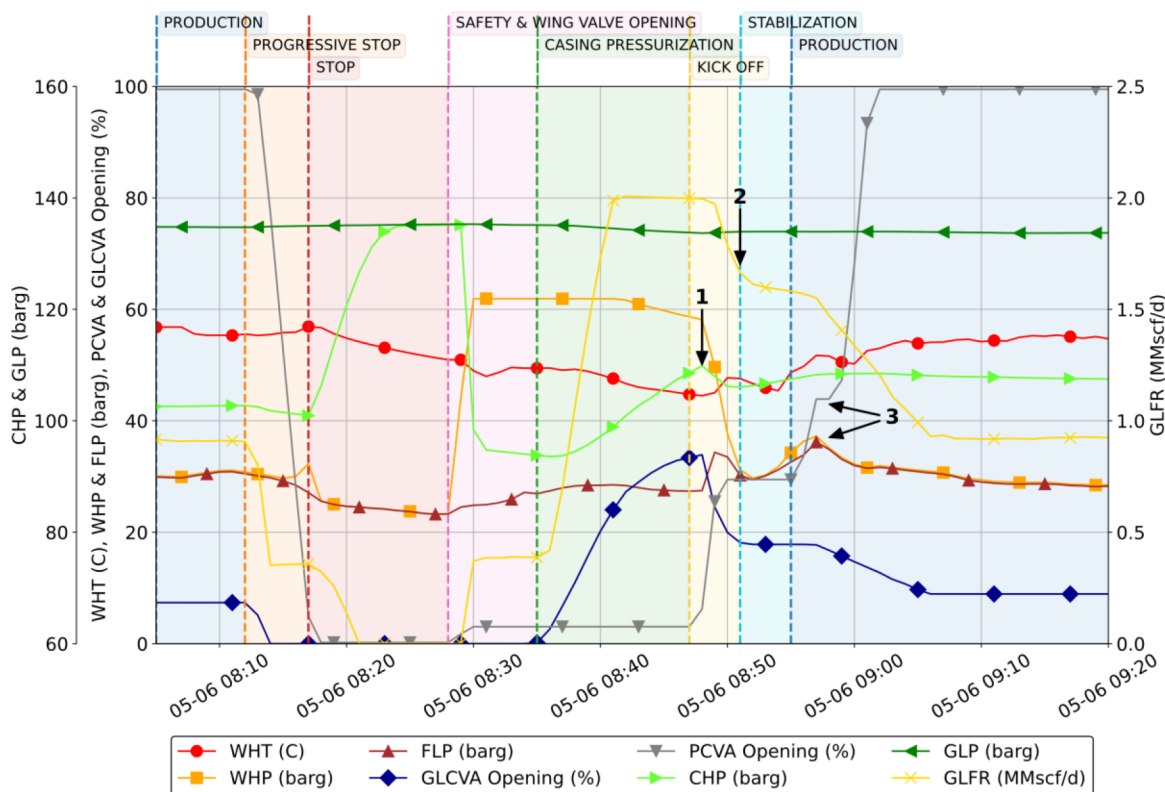


Figure 6—Well head parameters of gas-lifted well 8. FCW progressive stop followed by the “closed” start-up describe in Figure 2. Transparent colored bands indicate FCW states. Time axis: dd-mm hh:mm.

- **Production:** The well is regulated at its operating target using a fixed PCVA opening of 100% and GLFR = 0.9 MMscf/d.
- **Progressive stop:** Operator has clicked on the ‘Progressive stop’ button. GLCVA closes to 0%, and then PCVA ramps down to its heel position of 4% at a predefined speed.
- **Stop:** Safety valves PSSV and GLSSV are closed. PCVA closes to 0%. After safety valve closure, CHP equalizes to GLP while the gas-lift rate reduces to 0 MMscf/d, indicating a slight passing across GLCVA. WHP aligns with FLP, also suggesting minor passing across the PCVA.
- **Safety & wing valve opening:** Operator has clicked on the ‘start’ button and acknowledged that wing valves are open. This step aggregates steps 2-5 of Figure 2. Safety valves open and the production choke returns to its heel position. WHP = 61 barg, indicating a high presence of gas at the well head.
- **Casing pressurization:** CHP increases at a maximum rate of 1.2 bar/min to its target of 110 barg (see mark #1). Gas-lift rate peaks at 2 MMscf/d while PCVA remains at its heel position.
- **Kick-off:** The production choke opens to its target of 30% following a controlled ramp-up speed, and gas-lift is injected with a kick-off rate of 1.6 MMscf/d (see mark #2).
- **Stabilization:** PCVA and GLFR are held constant for 5 min at their respective targets.
- **Return to production:** After the stabilization timer elapses, PCVA and GLFR are ramped to their steady-state targets of 100% and 0.9 MMscf/d, respectively. At 8h57, FLP protection activates at

the high threshold of 35 barg, temporarily pausing the choke opening to avoid a safety trip (see mark #3).

A second example, illustrating a ‘progressive stop’ followed by an ‘open’ start-up, using FCW, is presented in Figure 7. The key distinction from the ‘closed’ start-up sequence is the inclusion of a ‘tubing depressurization’ step, during which the PCVA is opened to a fixed position prior to the ‘casing pressurization’ phase. This approach helps mitigate potential increases in flowline pressure that can occur when the production choke is opened after casing pressurization.

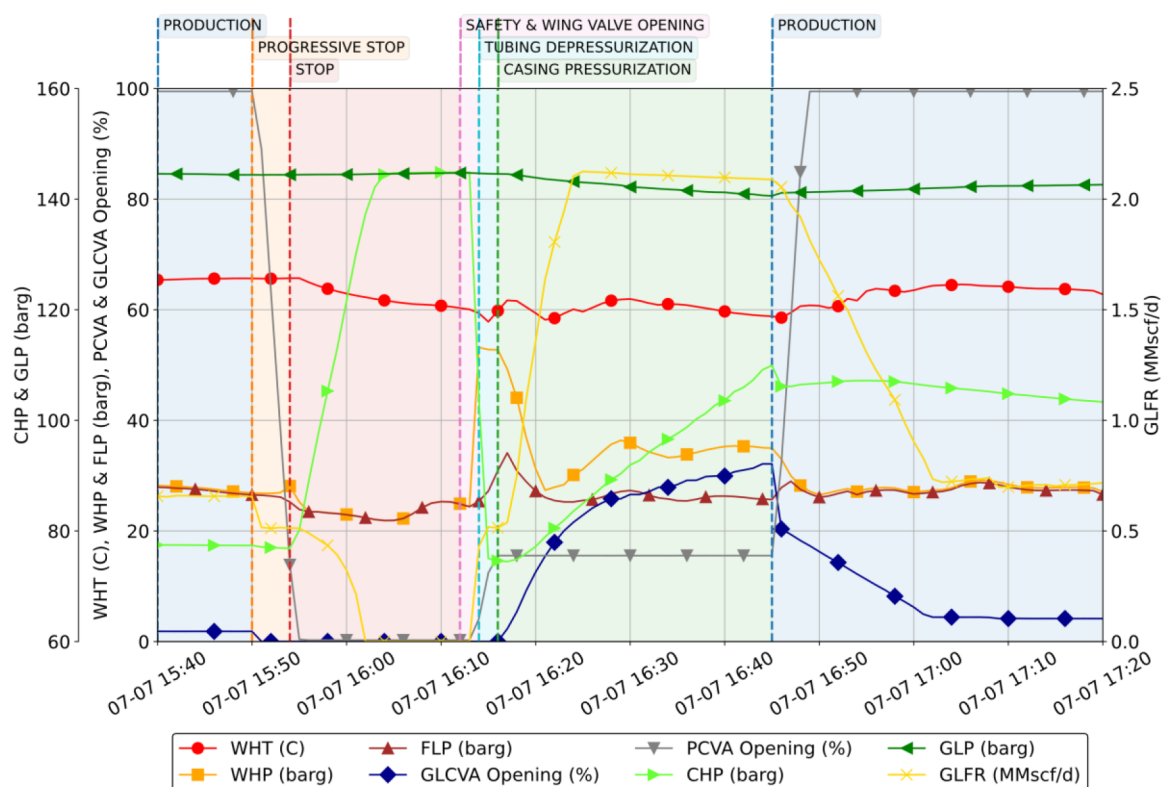


Figure 7—Well head parameters of gas-lifted well 8. FCW progressive stop followed by an FCW "open" start-up. Transparent colored bands indicate FCW states. Time axis: dd-mm hh:mm.

Figure 8 illustrates a manual stop and restart sequence. To shut-in the well, both the gas-lift and production chokes were closed instantaneously to 0%, without any ramp-down, while the safety valves remained open. A positive gas-lift rate observed with the GLCVA fully closed indicates leakage through the gas-lift choke. To restart, the PCVA was opened directly to 100%, and the GLCVA was set to a fixed position of 8%. During this phase, CHP gradually increased while the gas-lift rate declined from 1.5 MMscf/d to approximately 1.1 MMscf/d over three hours. Concurrently, GLP decreased from 160 barg to 150 barg. In this scenario, the gas-lift rate was not actively regulated to an optimal target but instead stabilized at a value dictated by the prevailing gas-lift pressure and the fixed GLCVA opening.

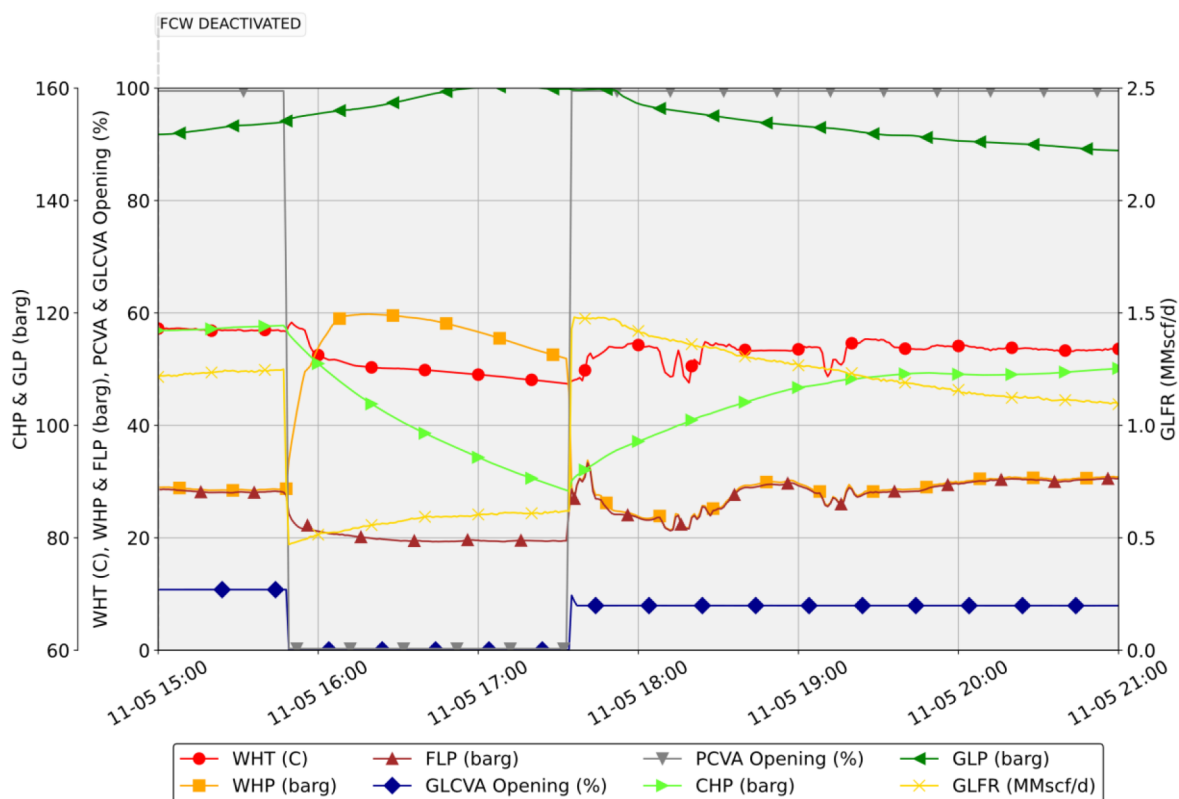


Figure 8—Well head parameters of gas-lifted well 8. Manual stop followed by a manual restart. FCW is deactivated. Time axis: dd-mm hh:mm.

These examples highlight the operational advantage of FCW: the well reaches stable CHP and gas-lift rate in approximately one hour, compared to nearly three hours with manual restart. However, for a fairer comparison, both methods should ideally be tested under identical shut-in durations to isolate the impact of the control strategy.

Stop and start-up of an ESP well

An example of a typical progressive stop followed by a start-up of an ESP well, using FCW, is shown in Figure 9.

- Production: The well operates at its target: WHP = 31 barg and fixed motor frequency = 41 Hz.
- Progressive stop: The operator has clicked on the 'Progressive stop' button. The VSD frequency ramps down to its minimum value of 38 Hz at a controlled deceleration rate, while the production choke continues to regulate at the operating target. A stop command is then sent to the VSD, shutting down the pump. Once the pump stops, the production choke closes to 0%.
- Waiting for ESP: The operator has clicked on the 'start' button and acknowledged that wing valves are open. The production choke opens to its heel position of 10%. The sequence then waits for ESP readiness, which includes no back spin, zero differential pressure across the pump and a minimum wait time of 15 minutes, as per the field operating philosophy and the pump protection principles.
- Tubing fill-up: A start command is sent to the pump, which starts at its start-up frequency (35 Hz). The frequency then ramps at a defined transition speed to the minimum value of 38 Hz, while the production choke regulates at the desired WHP target.
- Return to production: The frequency ramps-up at a predefined speed to the operating target of 41 Hz, with the production choke maintaining WHP control. Under these conditions, intake and

discharge pressures require approximately 1 hour to stabilize, resulting in active PCVA adjustments to maintain the WHP target.

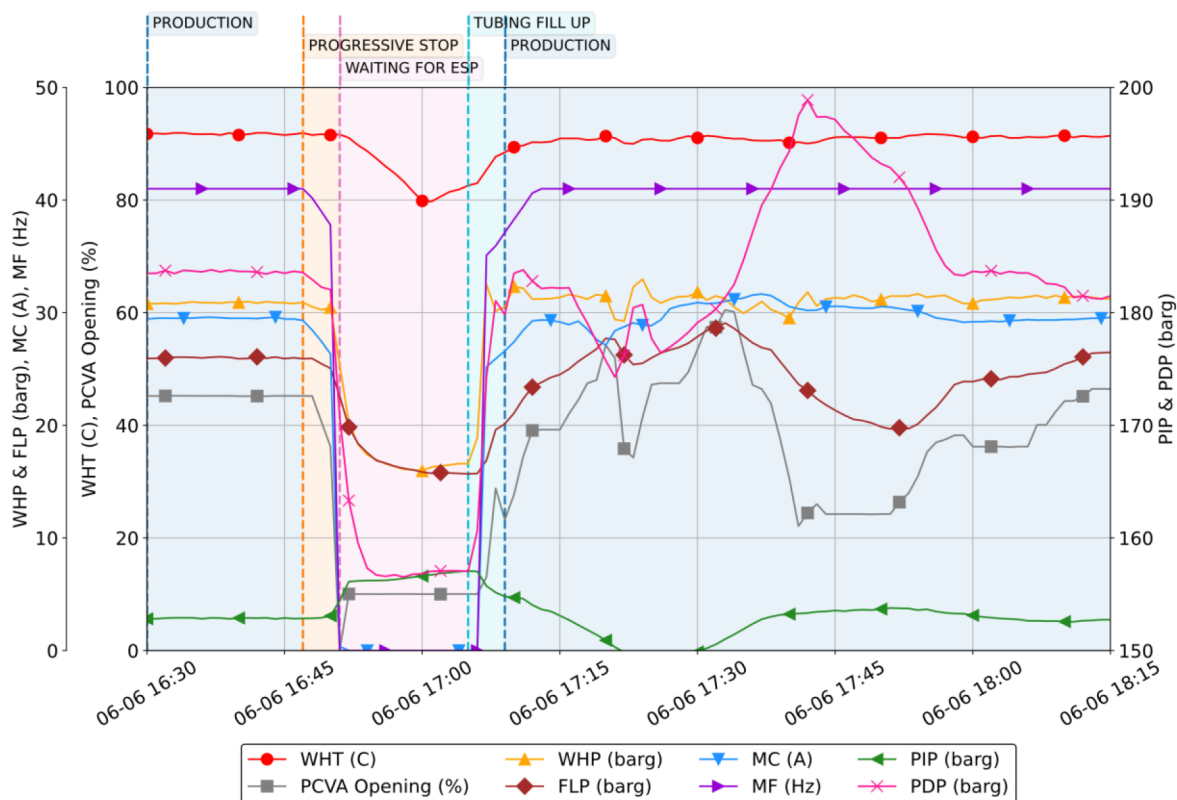


Figure 9—Well head and ESP parameters of ESP well 5. FCW progressive stop followed by an FCW start-up. Transparent colored bands indicate FCW states. Time axis: dd-mm hh:mm.

A manual stop and restart are shown in Figure 10, for comparison. To stop the well, the pump was shut down without ramping down its frequency, and the PCVA remained at a fixed opening. To restart, the VSD sets the pump at an initial frequency of 35 Hz, which is then manually adjusted to 40 Hz by the operator. This abrupt change caused a sharp spike in ESP discharge pressure, WHP, and motor current (not shown in the plot because the signal was not yet mapped from the VSD to the DCS). These sudden pressure surges led to a rapid increase in ESP load, which – if repeated – can impose excessive mechanical stress on the pump and reduce its long-term reliability. Discharge pressure and WHP remained unstable for several hours after restart.

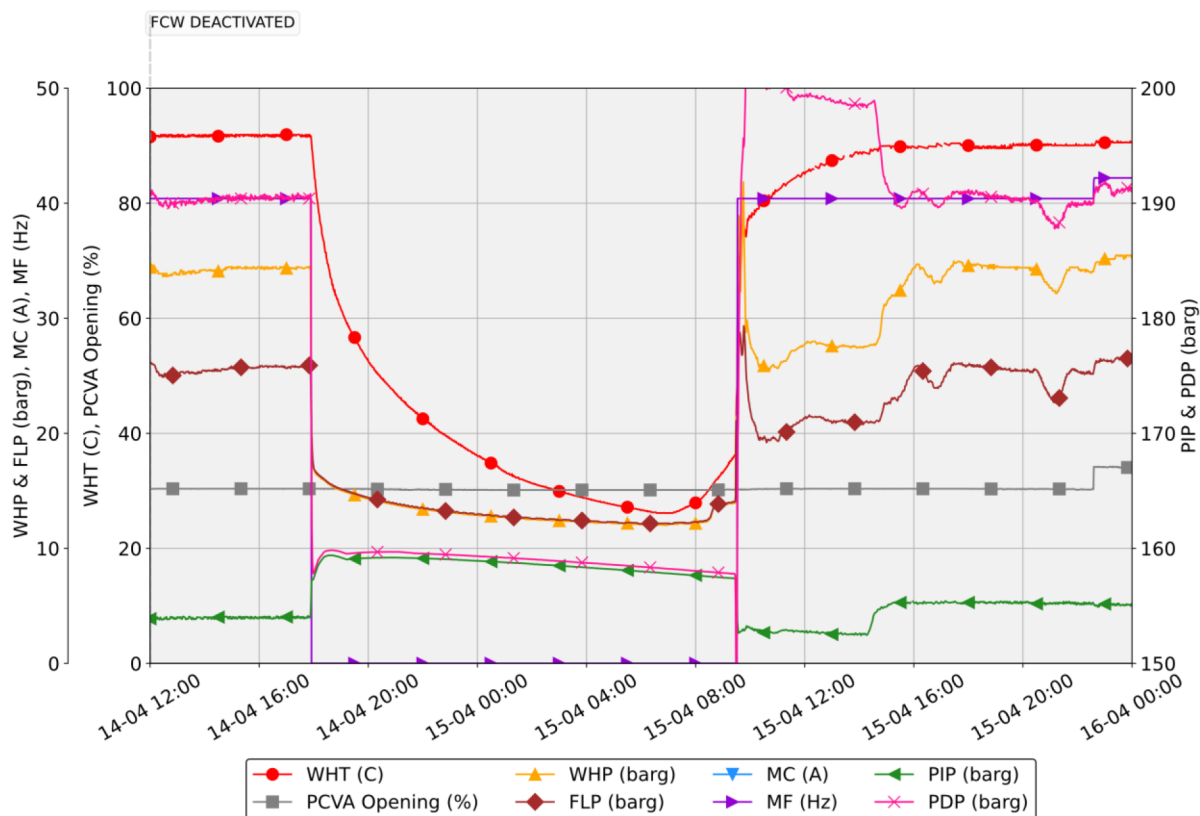


Figure 10—Well head and ESP parameters of ESP well 5. Manual stop followed by a manual restart. FCW is deactivated. Time axis: dd-mm hh:mm.

Production and gas-lift partial load shedding

To demonstrate the functionality of the multi-well module, the operator initiated a production partial load-shedding scenario by selecting the corresponding control button. This scenario involved one gas-lifted well and one ESP well, with the associated well signals shown in Figure 11. Upon activation, the production choke of gas-lifted well 8 – ranked as the first well to be load shed – was ramped down to a predefined operating opening of 10%, resulting in an increase in WHP above FLP, indicating a reduction in production rate. After a one-minute adjustable delay, the motor frequency of ESP well 9 – ranked second – was reduced to its minimum value of 38 Hz. Ten minutes later, the scenario was deactivated, and both wells returned to normal operation following the predefined restart sequence: well 8 first, followed by well 9 after the specified delay.

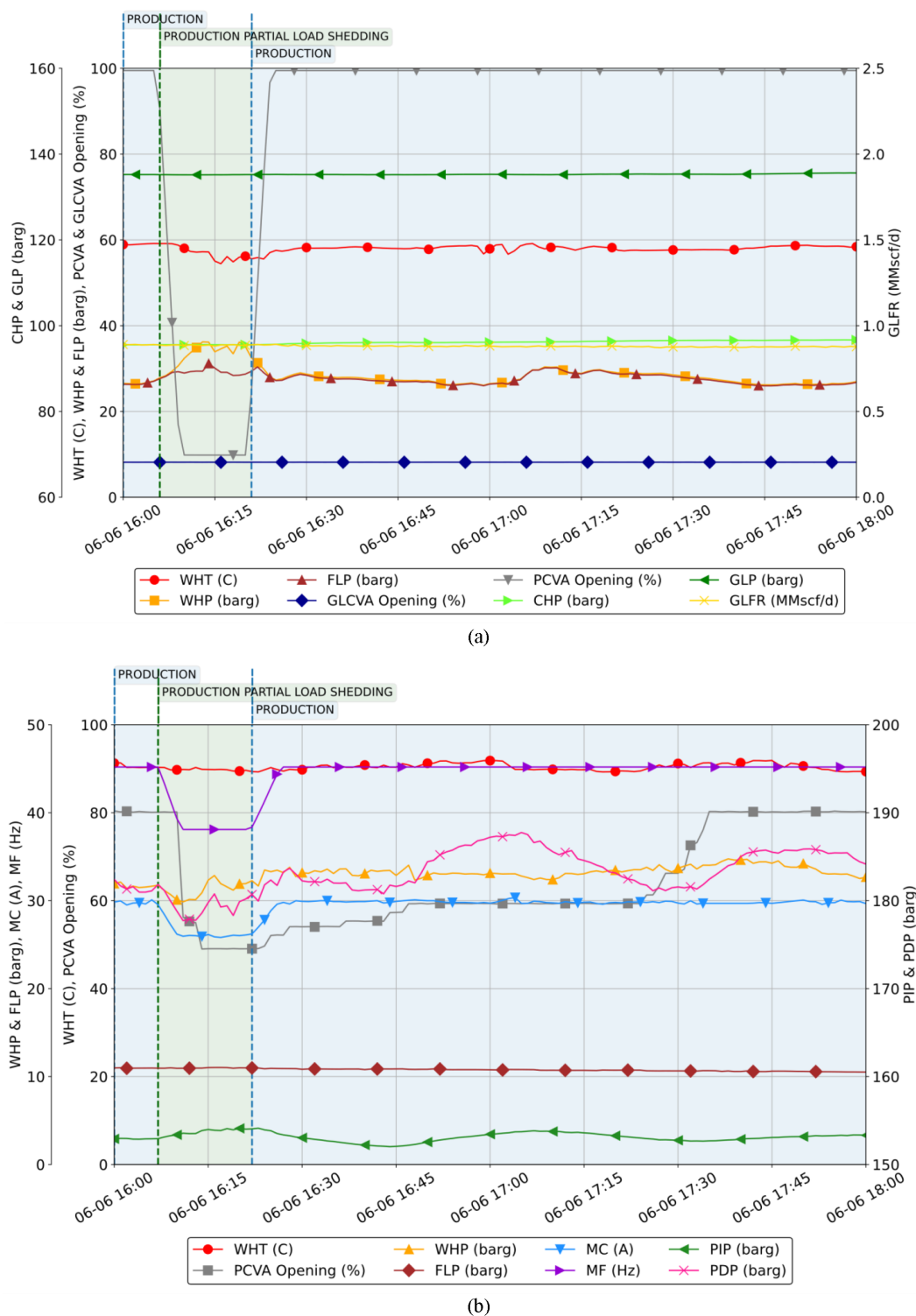


Figure 11—Well head and ESP parameters evolution during a FCW multi-well liquid partial load shedding scenario activated for gas-lifted well 8 and ESP well 9. Transparent colored bands indicate FCW states. Time axis: dd-mm hh:mm. (a) Gas-lifted well 8. (b) ESP well 9.

A second example is shown in Figure 12, where operators initiated another load-shedding scenario activation – this time a gas-lift partial load shedding involving two gas-lifted wells. In this case, well 3, ranked first, had its gas-lift rate reduced to 0.5 MMscf/d. After a one-minute adjustable delay, well 8, ranked second, was adjusted to its designated load-shedding value of 0.55 MMscf/d. As the previous scenario, the module restored both wells to their normal operating conditions once the scenario was deactivated.

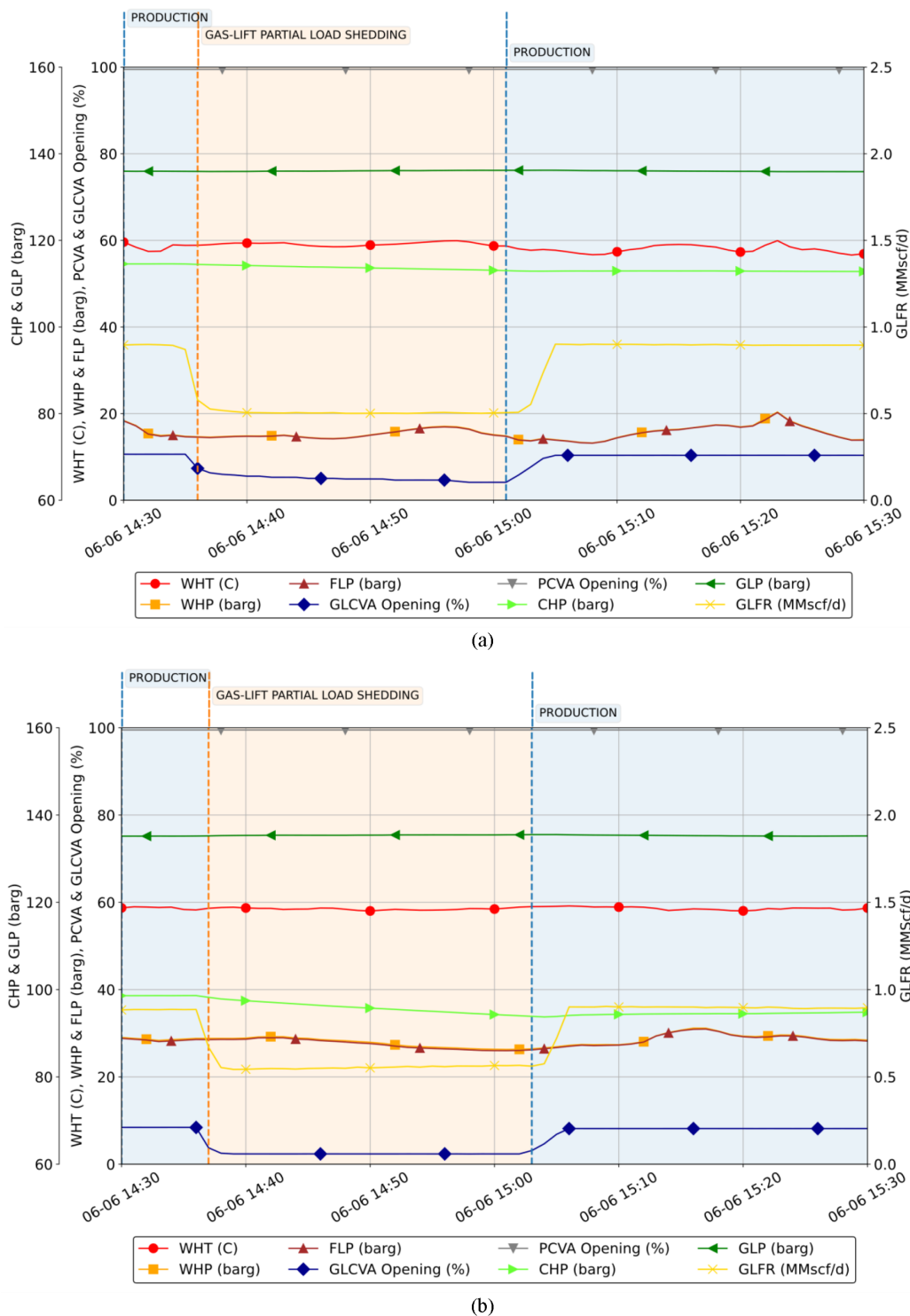


Figure 12—Well head parameters evolution during a FCW multi-well gas-lift partial load shedding scenario activated for gas-lifted wells 3 and 8. Transparent colored bands indicate FCW states. Time axis: dd-mm hh:mm. (a) Gas-lifted well 3. (b) Gas-lifted well 8.

Lessons learned and challenges

This section summarizes the key lessons learned and challenges encountered during the deployment of the FCW system, along with considerations for a future extension to other wells.

Clear and precise functional analyses were critical to minimize ambiguity and avoid rework during logic implementation by the ICSS vendor. Several typical acceptance tests (TAT) were conducted remotely before

the factory acceptance test (FAT) to validate core functionalities. However, remote debugging and logic validation via screen sharing proved inefficient and time-consuming, ultimately extending the FAT timeline and offsetting the initial cost savings of remote execution. For future deployments, conducting the TAT on-site should be considered to streamline validation and improve overall efficiency. Finally, to avoid delays during the Site Acceptance Test (SAT), we found it highly beneficial to conduct a site survey focused on instrumentation, a few weeks in advance. The FAT, which brings the project team together, offers an ideal opportunity to coordinate this survey efficiently.

Several technical and operational challenges were encountered. Data quality issues, such as the failure of an ESP downhole sensor and intermittent server connectivity affecting the data historian disturbed the system reliability and the efficiency of its monitoring. Instrumentation issues, such as oversized, partially obstructed or chokes failing to open/close properly prevented proper system configuration in some cases. Finally, operator adoption required significant effort, with multiple training sessions organized to address change management and build trust in the tool. To support field operations, permanent remote monitoring by an FCW expert is also recommended.

Scaling the FCW logic within the same field is relatively straightforward, typically involving duplication of existing sequences, integration of new valve typicals, and a brief testing phase to ensure error-free implementation – requiring about half a day per well. However, adapting the logic to other fields introduces complexity due to ICSS integration constraints. Even with the same vendor, differences in software versions or hardware generations, as well as varying operational philosophies may require logic adjustments. Transitioning to a different ICSS vendor necessitates complete redevelopment of the logic, significantly increasing the workload. Additionally, variations in well instrumentation and differences in processing infrastructure, such as gathering stations, may require tailored modifications to both mono-well and multi-well modules.

Conclusion

The purpose of this paper was to present the pilot results of the implementation of advanced control sequences – referred as the Full Control of Well (FCW) – within the ICSS of a Middle East onshore field, aimed at automating nine artificially lifted producer wells (gas-lift and ESP). The sequences covered the full operational lifecycle: start-up, stop, regulation at a defined target and load-shedding in response to process constraints.

As wells were not equipped with in-line multi-phase flowmeters to measure their production in real time, a fully quantitative assessment of the solution – specifically in terms of net oil gain – remains challenging. Alternative methods, such as virtual flowmeter or back-allocation of the field measured production using periodic well tests, proved unreliable, as the expected gain falls within the uncertainty range of these estimates. Nonetheless, the pilot demonstrated two clear operational advantages. First, well control is now fully automated, allowing operators to manage complex operations – such as start-up or load shedding – with the use of a few buttons. Second, start-up performance improved: gas-lifted wells reached stabilization several hours faster than with manual control, and ESP wells exhibited smoother transitions in critical signals like discharge pressure. The system proved reliable, remaining active for more than 75% of the total pilot duration, and more than 80% of the wells' flowing time. Downtime was mainly due to technical issues, such as failures in the hydraulic pump controlling the SSSV, which triggered well trips and deactivated FCW.

The system requires configuration of a large number (>50) of input parameters per well – targets, ramps, PID parameters, timers, and limits – but most were set once and remained unchanged throughout the pilot. Some parameters typically only need updating in response to significant changes in well behavior, following a well intervention for instance. In contrast, targets, such as gas-lift rate or motor frequency were adjusted regularly by operators, often daily, based on Production Optimization team recommendations. Looking ahead, the system could be integrated with a network model to dynamically feed input parameters. However,

as of today, this represents a technical challenge as ICSSs are usually isolated from IT networks where these models are hosted, for cybersecurity reasons.

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Nomenclature

APC	Advanced Process Control
BHP	Bottom Hole Pressure
CDS	Central Degassing Station
CHP	Casing Head Pressure
DCS	Distributed Control System
ESP	Electrical submersible pump
FAT	Factory Acceptance Test
FCW	Full Control of Well
FLP	Production Flowline Pressure
GLCVA	Gas-Lift Choke
GLFR	Gas-lift Flow Rate
GLP	Gas-lift (Flowline) Pressure
GLSSV	Gas-Lift Surface Safety Valve
GLWV	Gas-lift Wing Valve
HMI	Human Machine Interface
KPI	Key Performance Indicator
ICSS	Integrated Control and Safety System
LS	Load Shedding
PCVA	Production Choke
PDP	Pump Discharge Pressure
PIP	Pump Intake Pressure
PLC	Programmable Logic Controller
PDT	Pump Discharge Temperature
PSSV	Production Surface Safety Valve
PWV	Production Wing Valve
SAT	Site Acceptance Test
SSSV	Sub Surface Safety Valve
TAT	Typical Acceptance Test
VSD	Variable Speed Drive
WHP	Well Head Pressure
WHT	Well Head Temperature

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